

Program 6: Maven Project To Automate Web Servers Using Selenium Dependency (using Firefox)

Websites used for Automation:

1. Sauce Demo: <https://www.saucedemo.com/>
2. Automation Exercise: <https://automationexercise.com/products>
3. Practice Test Automation: <https://practicetestautomation.com/practice-test-login/>

Actions to be performed by the Automation task in the FireFox browser:

1. Login to the SauceDemo website
2. Search and add a product to the cart on AutomationExercise website
3. Login to the Practice Test Automation website
4. Demonstrate multi-tab browser automation using Selenium WebDriver

Create a Maven Project:

```
mvn archetype:generate -DgroupId=com.example -  
DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-  
quickstart -DinteractiveMode=false
```

(Make sure to create the projects for both website 2 and 3)

```
rakshith@rakshith-VirtualBox: ~/devops
rakshith@rakshith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp05 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] Parameter: basedir, Value: /home/rakshith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp05
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Old (1.x) Archetype in dir: /home/rakshith/devops/MyMavenSeleniumApp05
[INFO] BUILD SUCCESS
[INFO] Total time: 1.531 s
[INFO] Finished at: 2026-03-19T15:08:59+05:30
[INFO]
rakshith@rakshith-VirtualBox:~/devops$
```

Before moving into the next step check if the required driver is installed and it supports the current browser being used.

Add Dependencies in pom.xml:

```
Open pom.xml Save
~/Devops/MyMavenSeleniumApp05
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5   <groupId>com.example</groupId>
6   <artifactId>MyMavenSeleniumApp05</artifactId>
7   <packaging>jar</packaging>
8   <version>1.0-SNAPSHOT</version>
9   <name>MyMavenSeleniumApp05</name>
10  <url>http://maven.apache.org</url>
11  <dependencies>
12    <dependency>
13      <groupId>junit</groupId>
14      <artifactId>junit</artifactId>
15      <version>3.8.1</version>
16      <scope>test</scope>
17    </dependency>
18    <dependency>
19      <groupId>org.seleniumhq.selenium</groupId>
20      <artifactId>selenium-java</artifactId>
21      <version>4.41.0</version>
22      <scope>compile</scope>
23    </dependency>
24  </dependencies>
25  <!-- Build: Configuring plugins and build settings -->
26  <build>
27    <!-- Example: Maven Compiler Plugin to compile Java code -->
28    <plugin>
29      <groupId>org.apache.maven.plugins</groupId>
30      <artifactId>maven-compiler-plugin</artifactId>
31      <version>3.8.1</version>
32    </plugin>
33  </build>
34  </project>
35
```

```
Open  pom.xml  Save  ~/Devops/MyMavenSeleniumApp05
60 <plugin>
61     <groupId>org.apache.maven.plugins</groupId>
62
63     <artifactId>maven-jar-plugin</artifactId>
64     <version>3.0.1</version>
65     <configuration>
66         <archive>
67             <manifestEntries>
68                 <Main-Class>com.example.App</Main-Class>
69             </manifestEntries>
70         </archive>
71     </configuration>
72
73
74 </plugin>
75 <plugin>
76     <groupId>org.apache.maven.plugins</groupId>
77     <artifactId>maven-shade-plugin</artifactId>
78     <version>3.5.0</version>
79     <executions>
80         <execution>
81             <phase>package</phase>
82             <goals>
83                 <goal>shade</goal>
84             </goals>
85             <configuration>
86                 <transformers>
87                     <transformer implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">
88                         <mainClass>com.example.App</mainClass>
89                     </transformer>
90                 </transformers>
91             </configuration>
92         </execution>
93     </executions>
94 </plugin>
95
96 </plugins>
97 </build>
98 </project>
```

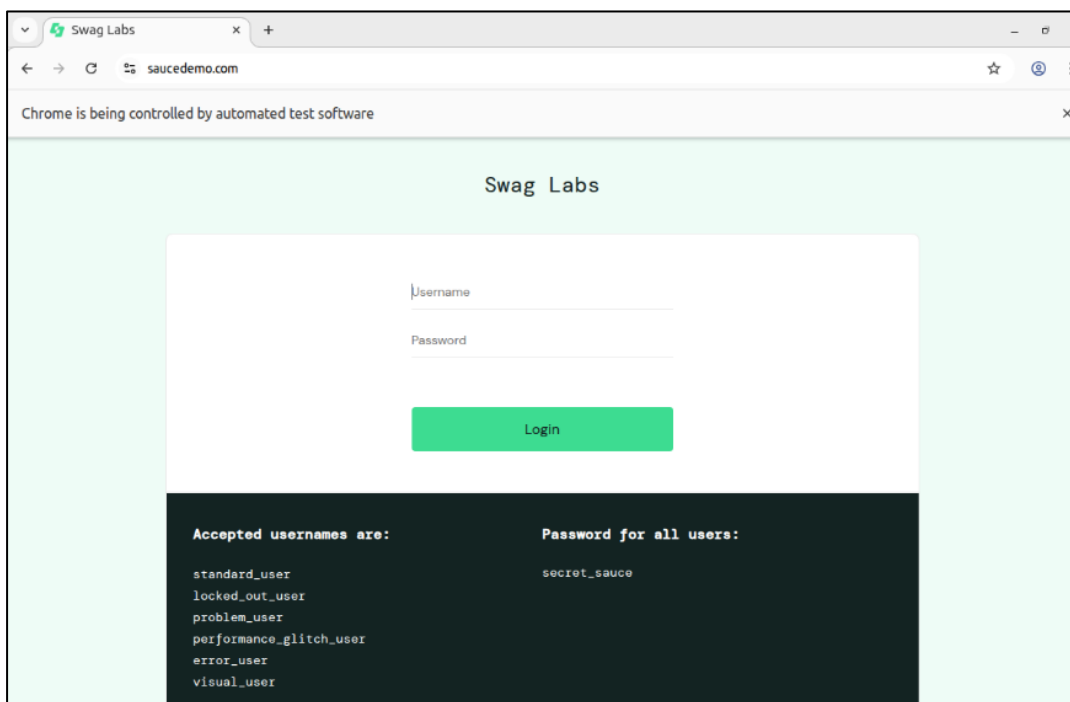
Edit App.java:

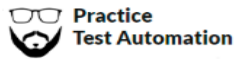
```
Open  *App.java  Save  ~/Devops/MyMavenSeleniumApp05/src/main/java/com/example
1 package com.example;
2 import org.openqa.selenium.By;
3 import org.openqa.selenium.WebDriver;
4 import org.openqa.selenium.WebElement;
5 import org.openqa.selenium.WindowType;
6 import org.openqa.selenium.firefox.FirefoxDriver;
7 import org.openqa.selenium.firefox.FirefoxOptions;
8 import org.openqa.selenium.interactions.Actions;
9 import org.openqa.selenium.support.ui.ExpectedConditions;
10 import org.openqa.selenium.support.ui.WebDriverWait;
11 import java.time.Duration;
12 public class App {
13
14     public static void main(String[] args) throws InterruptedException {
15
16         FirefoxOptions options = new FirefoxOptions();
17
18         WebDriver driver = new FirefoxDriver(options);
19
20         driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));
21
22         WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));
23
24         Actions actions = new Actions(driver);
25
26         driver.get("https://www.saucedemo.com/");
27
28         driver.findElement(By.id("user-name")).sendKeys("standard_user");
29
30         driver.findElement(By.id("password")).sendKeys("secret_sauce");
31
32         driver.findElement(By.id("login-button")).click();
33
34         System.out.println("SauceDemo login successful");
35
36         driver.switchTo().newWindow(WindowType.TAB);
37
38         driver.get("https://automationexercise.com/products");
```

```
Open  [icon]  *App.java  Save  [icon]  [icon]  [icon]  [icon]
~/Devops/MyMavenSeleniumApp05/src/main/java/com/example

50  ((org.openqa.selenium.JavascriptExecutor) driver)
51  .executeScript("arguments[0].scrollIntoView({block:'center'});", product);
52
53  ((org.openqa.selenium.JavascriptExecutor) driver)
54  .executeScript("arguments[0].click();", product);
55
56  WebElement viewCart = wait.until(
57      ExpectedConditions.elementToBeClickable(
58          By.cssSelector("#cartModal a[href='/view_cart']")
59      )
60  );
61
62  ((org.openqa.selenium.JavascriptExecutor) driver)
63  .executeScript("arguments[0].scrollIntoView(true);", viewCart);
64
65  ((org.openqa.selenium.JavascriptExecutor) driver)
66  .executeScript("arguments[0].click();", viewCart);
67
68  System.out.println("Automation Exercise product added to cart");
69
70  driver.switchTo().newWindow(WindowType.TAB);
71
72  driver.get("https://practicetestautomation.com/practice-test-login/");
73
74  Thread.sleep(2000);
75
76  driver.findElement(By.id("username")).sendKeys("student");
77
78  driver.findElement(By.id("password")).sendKeys("Password123");
79
80  driver.findElement(By.id("submit")).click();
81
82  System.out.println("Practice Test Automation login successful");
83
84  Thread.sleep(5000);
85
86  driver.quit();
87  }
88 }
```

Run the Application:





Logged In Successfully

Congratulations student. You successfully logged in!

[Log out](#)



[Home](#) [Shopping Cart](#)

[Proceed To Checkout](#)

Item	Description	Price	Quantity	Total
	Blue Top Women > Tops	Rs. 500	1	Rs. 500

